CM2014 Simulation methods in Medical Engineering
Spring 2025

Lennart Edsberg

Mathematical and Numerical background

Lecture 1

Model problems, first order Euler-methods, MATLAB

Literature: 1) ”Introduction to computation and modeling for differential equations”, (LE), Lennart Edsberg, Wiley, 2nd edition, textbook available at KTH bookstore, main campus, Valhallavägen and at Bokus
2) An Introduction to MATLAB, version 4.1 (2018), David Griffiths, Univ of Dundee, just google on the title and you have a free copy, 3) Text to Lab 1 and Lab 2 , 4) Handouts from lectures, 5) MATLAB-programs

Theory: The following parts (ordinary differential equations, ODE's) in textbook 1) are included in CM2014
Ch 1: Whole chapter, Ch 2: 2.1, 2.2, 2.3 (intro + example 2.1), 2.5 (example 2.8)
Ch 3: Intro, 3.1, 3.2, 3.3 intro, 3.3.1, (not pg 45-46), 3.3.2, 3.3.3, 3.3.4,3.3.5,3.4.1 (pg 56-57), 3.4.2 (intro + example 3.6), 3.5
Ch 4; 4.1 (intro + example 4.1), 4.2 (intro + 4.2.1)
The rest of the book is not compulsory for this part of the course but the contents are valuable for the rest of the course: Finite Elements, Simulation of fluid mechanics, Molecular simulations, etc. 4) Handouts are included.

Labs: Labs to be done in MATLAB in groups of two students in each group.
Lab 1 (CL1-GA) and Lab 2 (CL1-GB) are compulsory. Lab 1 is about an initial value problem for the harmonic oscillator, Lab 2 is about an initial value problem for the kinetics of a biochemical reaction.
The Labs are **graded**.
* To get a **pass**, the students of the group must complete the lab assignments with satisfaction, meaning that the working MATLAB-programs and the results (graphs) must be submitted as pdf-files via Canvas
* To get a **pass with Excellence** the students must **also** produce a well-written report that demonstrates a deep understanding of the theoretical background by relating to **several relevant** key concepts taught. The report should be submitted together with the working MATLAB-programs and results as pdf-files via Canvas
* Last day for submission of the two Labs is March 3rd 2024.
* Lab guidance and help available at scheduled labs. At other times, ask questions to edsberg@kth.se

Prerequisites for this course are knowledge in ordinary differential equations (ODE's), linear algebra and programming in MATLAB.

Important numerical concepts are *accuracy, stability and efficiency*

The following ODE-problems will be used as *model problems (*MP*)* to illustrate mathematical and numerical concepts and phenomena

MP1) Exponential growth/decay, a scalar linear first order ODE: du/dt = lambda u(LE, pg 2)

MP2) The harmonic oscillator, a scalar linear second order ODE: d^2 x/dt^2 + (omega^2) x =0

MP3) The advection-diffusion equation, a scalar linear second order ODE: –d^2 u/dx^2 + a du/dx = f(x)

Problem 1) and 2) will be studied as an *initial value problem* (IVP), i.e.
1) u(0)=u0, initial value u0 is given at t=0
2) x(0)=x0, dx/dt (0)=v0, inital values x0 and v0 are given, (x denotes position, v velocity) at t=0

Problem 3) will be studied as a *boundary value problem* (BVP), i.e.
3) u(0)=alfa, u(1)=beta, boundary values alfa and beta are given at the boundaries x=0 and x=1

Model problem MP1 is a differential eqn (D-eqn)

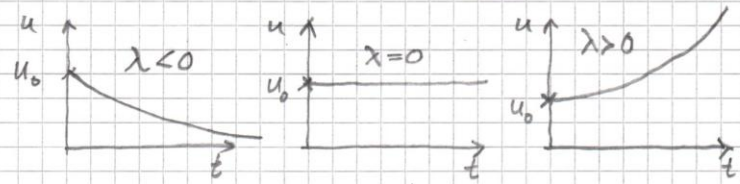
$\frac{du}{dt} = \lambda u$ u dependent variable, t independent variable, λ parameter $\Rightarrow u(t) = Ce^{\lambda t}$ C arbitrar constant

∞ many solutions, choose <u>one</u> of them by giving an initial condition, $u(0) = u_0$

→ IVP: $\frac{du}{dt} = \lambda u,\ u(0) = u_0 \Rightarrow u(t) = u_0 e^{\lambda t}$ (1

Initial value Problem

a, Assume λ is a <u>real</u> number



$u(t)$ decreasing — u constant — $u(t)$ increasing

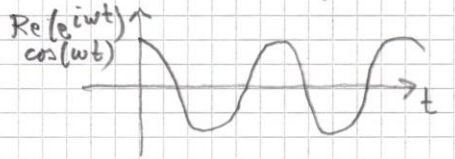
$u(t) \to 0$ as $t \to \infty$ — $u(t) \to \infty$ as $t \to \infty$

↖ $u(t)$ bounded (stable) — $u(t)$ unbounded (unstable)

b, Assume λ is a <u>complex</u> number $\lambda = \mu + i\omega$

$u(t) = u_0 e^{(\mu+i\omega)t} = u_0 e^{\mu t} e^{i\omega t}$ (2)

$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$ bounded (stable) for all ω



See also exercises 1.2.1, 1.2.2, 1.2.3 in LE

(1) is stable if $\lambda \le 0$

(2) is stable if $\mathrm{Re}(\lambda) \le 0$

Model problem 1 generalized to a system of ODE's on component form

$$\frac{du_1}{dt} = a_{11}u_1 + a_{12}u_2, \quad u_1(0) = u_{10}$$
$$\frac{du_2}{dt} = a_{21}u_1 + a_{22}u_2, \quad u_2(0) = u_{20} \qquad (1)$$

can also be written on matrix form

$$\frac{d\bar{u}}{dt} = A\bar{u}, \quad \bar{u}(0) = \bar{u}_0 \qquad (1)$$

where $\bar{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $\bar{u}_0 = \begin{pmatrix} u_{10} \\ u_{20} \end{pmatrix}$

2×2 matrix

System (1) is linear with constant coefficients

General nonlinear system of ODE's

$$\frac{d\bar{u}}{dt} = \bar{f}(t, \bar{u}), \quad \bar{u}(0) = \bar{u}_0 \qquad (2)$$

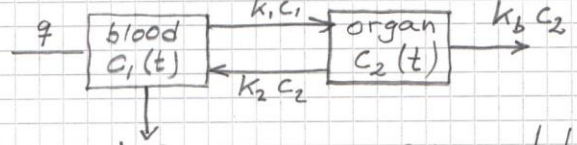
(2) is the standard formulation and starting point for using MATLAB's ODE-solvers, e.g. ode23, ode45

In case there is only one nonlinear ODE, the system (2) turns into a nonlinear scalar ODE

$$\frac{du}{dt} = f(t, u), \quad u(0) = u_0 \qquad (3)$$

Examples of IVPs

1) Drug kinetics (example 2.8 in LE)



MATLABs ode45 used.

component form

$$\frac{dC_1}{dt} = -k_1 C_1 + k_2 C_2 - k_e C_1 + q \quad , C_1(0) = 0$$

drug.m + fdrug.m

$$\frac{dC_2}{dt} = k_1 C_1 - k_2 C_2 - k_b C_2 \quad , C_2(0) = 0$$

$$\frac{d\bar{C}}{dt} = \begin{pmatrix} -k_1 - k_e & k_2 \\ k_1 & -k_2 - k_b \end{pmatrix} \bar{C} + \begin{pmatrix} q \\ 0 \end{pmatrix}, \quad \bar{C}(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

matrix form

↑ driving function

2) Van der Pol oscillator (example 3.3 in LE)

vdp.m + fvdp.m

$$\frac{d^2y}{dt^2} + \varepsilon (y^2 - 1) \frac{dy}{dt} + y = 0, \quad y(0) = 1 \quad \frac{dy}{dt}(0) = 0$$

Nonlinear scalar second order ODE

Let $y = u_1$, $\frac{dy}{dt} = u_2$

trick:

$$\frac{du_1}{dt} = u_2 \qquad u_1(0) = 1$$

$$\frac{du_2}{dt} = -\varepsilon (u_1^2 - 1) u_2 - u_1 \qquad u_2(0) = 0$$

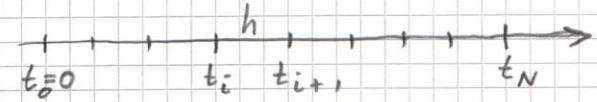
Nonlinear system of first order ODEs

* When $\varepsilon = 0$ we have the harmonic oscillator (for $w = 1$)

Numerical methods and MATLAB

Methods based on difference approximations (LE3) lead to difference equations (Δ-eqns)

1) Discretize the t-variable



(1)

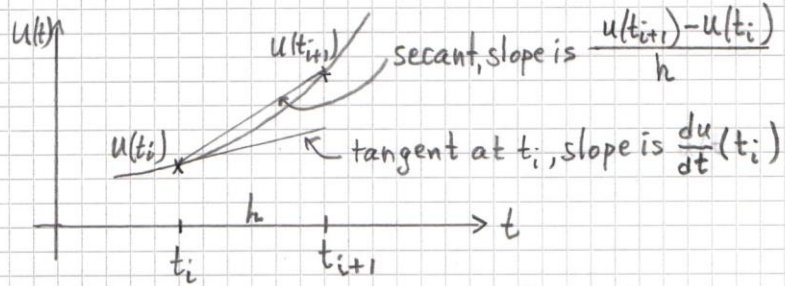
$t_i = ih$, $i = 0, 1, \ldots, N$, h is the stepsize

2) Approximate the derivative by a difference quotient

$$\dot{u}(t_i) = \frac{du}{dt}(t_i) \approx \frac{u(t_{i+1}) - u(t_i)}{h} \qquad (2)$$

This approximation is based on Taylor's formula:

Graphical illustration

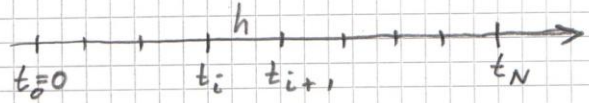


the secant slope is an approximation of the tangent slope, better the smaller h is

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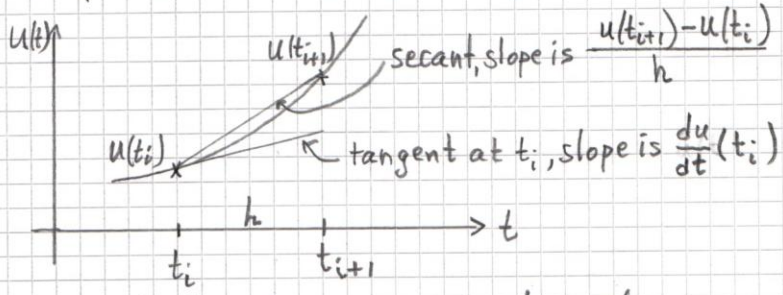
This approximation is based on Taylor's formula:

$$u(t_{i+1}) = u(t_i + h) = u(t_i) + h\dot{u}(t_i) + \frac{h^2}{2}\ddot{u}(\xi)$$

$$\Rightarrow \frac{u(t_{i+1}) - u(t_i)}{h} = \dot{u}(t_i) + \underbrace{\frac{h}{2}\ddot{u}(\xi)}_{O(h)} \quad \xi \in [t_i, t_{i+1}]$$

$O(h)$ meaning $\lim_{h\to 0} \frac{O(h)}{h} = C$

Graphical illustration $O(h) \approx Ch$



the secant slope is an approximation of the tangent slope, better the smaller h is

3, Apply (2) on $\frac{du}{dt} = \lambda u$, $u(0) = u_0$

At $t = t_i$: $\frac{u(t_{i+1}) - u(t_i)}{h} + O(h) = \lambda u(t_i)$

Skip the $O(h)$ - term and we obtain an approximation of the ODE with $u_i \approx u(t_i)$

$$\frac{u_{i+1} - u_i}{h} = \lambda u_i \quad , \; u_0 \text{ given}$$

$$\begin{cases} u_{i+1} = u_i + h\lambda u_i \\ t_{i+1} = t_i + h \end{cases}$$ Explicit Euler method

Example : Let $\lambda = -1$, $u_0 = 1$, $h = 0.1$

$$u_i = (1+h\lambda)\, u_{i-1} = (1+h\lambda)^2 u_{i-2} = \dots = (1+h\lambda)^i u_0$$

MATLAB-program euler1 tested with $h = 0.1, 1, 2, 3$

The sequence is bounded (stable) if

$$|1 + h\lambda| \leq 1 \qquad \text{assume } \lambda \text{ is real, } h > 0$$

Hence D-eqn $\dot{u} = \lambda u$, $u(0) = 1$ stable if $\lambda \leq 0$

Δ-eqn $u_{i+1} = u_i + h\lambda u_i$, $u_0 = 1$ stable if $-2 \leq h\lambda \leq 0$

3) Apply (2) on $\frac{du}{dt} = \lambda u$, $u(0) = u_0$

At $t = t_i$: $\frac{u(t_{i+1}) - u(t_i)}{h} + O(h) = \lambda u(t_i)$

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Example : Let $\lambda = -1$, $u_0 = 1$, $h = 0.1$

$u_1 = u_0 + h\lambda u_0 = 1 + 0.1 \cdot (-1) \cdot 1 = 0.9$

$t_1 = t_0 + h = 0 + 0.1 = 0.1$

$u_2 = u_1 + h\lambda u_1 = 0.9 + 0.1 \cdot (-1) \cdot 0.9 = 0.81$

$t_2 = t_1 + h = 0.1 + 0.1 = 0.2$

$\vdots$

$u_i = (1 + h\lambda) u_{i-1} = (1+h\lambda)^2 u_{i-2} = \ldots = (1+h\lambda)^i u_0$

MATLAB-program euler1 tested with $h = 0.1, 1, 2, 3$

The sequence is bounded (stable) if

$|1 + h\lambda| \leq 1$ assume λ is real, $h > 0$

$-1 \leq 1 + h\lambda \leq 1$

$h\lambda \geq -2$, $h\lambda \leq 0 \Rightarrow \lambda \leq 0$

$\Downarrow$

$-h\lambda \leq 2 \Rightarrow h \leq \frac{2}{-\lambda} = h_{max}$

$-1 \leq 1 + h\lambda \leq 1$

not ok if $\lambda > 0$

$\Rightarrow$ unstable if $\lambda > 0$

Hence D-eqn $\dot{u} = \lambda u$, $u(0) = 1$ stable if $\lambda \leq 0$

Δ-eqn $u_{i+1} = u_i + h\lambda u_i$, $u_0 = 1$ stable if $-2 \leq h\lambda \leq 0$

What about accuracy for Explicit Euler?

MATLAB-program euler2 computes solution of

$\frac{du}{dt} = \lambda u,\ u(0) = u_0,\ t \in [0, t_N]$ $\lambda = -1,\ u_0 = 1,\ t_N = 1$

with $u_{i+1} = u_i + h\lambda u_i$, $u_0 = 1$, $i = 0, 1, \ldots, N-1$, $hN = 1$

for $N = 10, 20, 40, 80, 160, 320$, $h = 1/N$

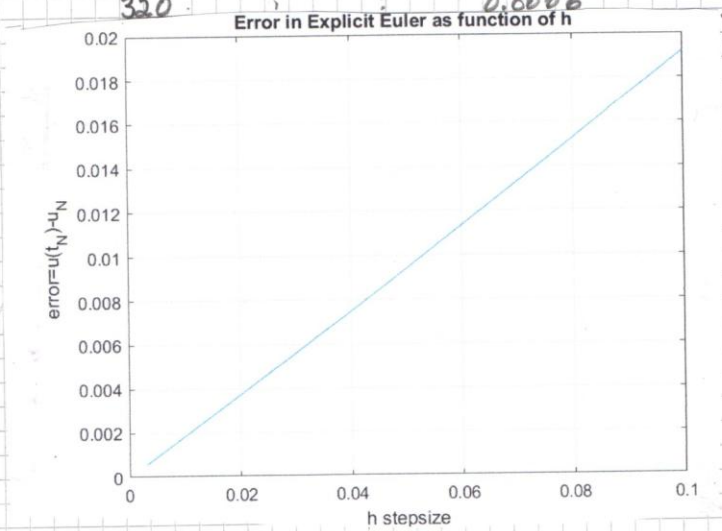
$u_N \approx u(t_N) = e^{-t_N} = e^{-1} = 0.3679$

Result

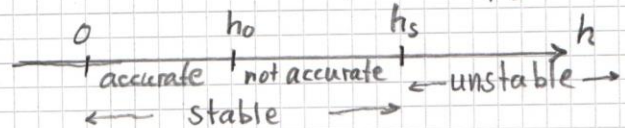
N	h	u_N	error $= e^{-1} - u_N$
10	0.1	0.3487	0.0192
20	0.05	0.3585	0.0094
40	0.025	0.3632	0.0046
80	⋮	⋮	0.0023
160	⋮	⋮	0.0012
320	⋮	⋮	0.0006

error $\approx ch$

$h = h^1$ ← first order method



See also LE Example 3.4, pg 44-45



The derivative approximation used in Explicit Euler:

① $\frac{du}{dt}(t_i) = \frac{u(t_{i+1}) - u(t_i)}{h} + O(h)$ forward diff

We can also use

② $\frac{du}{dt}(t_i) = \frac{u(t_i) - u(t_{i-1})}{h} + O(h)$ backward diff

③ $\frac{du}{dt}(t_i) = \frac{u(t_{i+1}) - u(t_{i-1})}{2h} + O(h^2)$ central diff

For the second derivative:

④ $\frac{d^2u}{dt^2}(t_i) = \frac{u(t_{i+1}) - 2u(t_i) + u(t_{i-1})}{h^2} + O(h^2)$

Apply ② on the MP1:

At $t = t_i$: $\frac{u(t_i) - u(t_{i-1})}{h} + O(h) = \lambda u(t_i)$

Skip the $O(h)$-term

$$\frac{u_i - u_{i-1}}{h} = \lambda u_i \quad , u_0 \text{ given}$$

$$\begin{cases} u_i = u_{i-1} + h\lambda u_i \\ t_i = t_{i-1} + h \end{cases}$$ Implicit Euler IE

Implicit since u_i occurs in both sides $\Rightarrow$ we must solve an equation w.r.t. u_i

$$u_i = \frac{1}{1 - h\lambda} u_{i-1}$$

Example: $\lambda = -1$, $h = 0.1$, $u_0 = 1$

$u_1 = \frac{1}{1.1} \cdot 1 \approx 0.9$

$u_2 = \frac{1}{1.1} \cdot 0.9 \approx 0.81$

$\vdots$

$u_i = \frac{1}{(1 - h\lambda)^i} u_0$

MATLAB-program eulerimp

$$u_i = \frac{1}{(1-h\lambda)^i}\, u_0 \quad , u_0 = 1$$

Stable if $|1-h\lambda| \geq 1$, $h>0$

$\lambda \leq 0$: stable for all h !

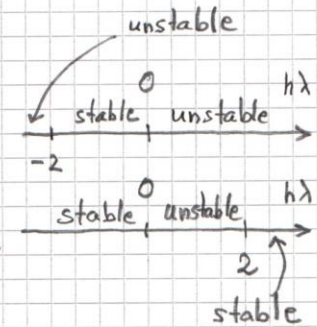
$\lambda > 0$: unstable if $|1-h\lambda| < 1$

$$-1 < 1-h\lambda < 1$$

$h\lambda < 2 \Rightarrow 0 < h < \frac{2}{\lambda}$ for instability

Hence we have the following stability results for EE and IE applied to MP1 :

D-eqn / Δ-eqn	$\lambda \leq 0$ stable	$\lambda > 0$ unstable
EE	stable if $-2 < h\lambda < 0$	unstable for all $h>0$
IE	stable for all $h>0$	unstable if $0 < h\lambda < 2$



Both methods are first order in accuracy, i.e.

$$u(t_i) - u_i = O(h)$$

Poor accuracy (small steps needed)

Better accuracy needed !

③ Central difference does not help. Stability problems. See MATLAB-program central .

Euler's explicit method applied to MPI

$$\frac{du}{dt} = \lambda u, \quad u(0) = u_0 \qquad IVP$$

$$\begin{cases} u_{i+1} = u_i + h\lambda u_i, & u_0 = \text{given value} \\ t_{i+1} = t_i + h, & t_0 = 0 \end{cases}$$

For a nonlinear scalar ODE

$$\frac{du}{dt} = f(t,u), \quad u(0) = u_0$$

$$\begin{cases} u_{i+1} = u_i + hf(t_i, u_i), & u_0 \text{ given} \\ t_{i+1} = t_i + h, & t_0 = 0 \end{cases}$$

For a general nonlinear system of ODE's

$$\frac{d\bar{u}}{dt} = \bar{f}(t,\bar{u}), \quad \bar{u}(0) = \bar{u}_0$$

$$\begin{cases} \bar{u}_{i+1} = \bar{u}_i + h\bar{f}(t_i, \bar{u}_i), & \bar{u}_0 \text{ given} \\ t_{i+1} = t_i + h, & t_0 = 0 \end{cases}$$

In Lab 1 we solve the harmonic oscillator

$$\frac{d^2x}{dt^2} + w^2 x = 0, \quad x(0) = 1, \quad \frac{dx}{dt}(0) = 0$$

with some different methods based on difference approximations.

First, rewrite the 2nd order ODE as a system: $u_1 = x$, $u_2 = \frac{dx}{dt}$ $\Rightarrow$

$$\begin{cases} \frac{du_1}{dt} = u_2, & u_1(0) = 1 \\ \frac{du_2}{dt} = -w^2 u_1, & u_2(0) = 0 \end{cases}$$

component form

On matrix form:

$$\frac{d\bar{u}}{dt} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}\bar{u}\,, \quad \bar{u}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Let the parameter $w = 2$ and stepsize $h = 0.1$

Euler's explicit method

$$\bar{u}_{i+1} = \bar{u}_i + h\begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}\bar{u}_i\,, \quad \bar{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

with numbers inserted

$$\bar{u}_1 = \bar{u}_0 + h\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\bar{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0.1\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -0.4 \end{pmatrix}$$

$$\bar{u}_2 = \bar{u}_1 + h\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\bar{u}_1 = \begin{pmatrix} 1 \\ -0.4 \end{pmatrix} + 0.1\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -0.4 \end{pmatrix} = \begin{pmatrix} 0.96 \\ -0.8 \end{pmatrix}$$

MATLAB-code in lab1demo (skip variable E)

```
y = [1, 0]';  t = 0;          initial value
A = [0 1; -omega^2 0];        matrix A
Y = [y']; T = [t];
for k = 1:N
    y = y + h*A*y;            Euler's method
    Y = [Y; y'];              recursive collection of y=ū_i
    t = t + h;
    T = [T; t];
end
plot(T, Y, '*')
```